

1 / 20 | Ages 5-6 | 10 minutes

Two clues, one animal

What we are practising

Eliminate choices that fail either clue.

What you need

Paper / Pencil

Name the four choices aloud.

Before you begin

Draw or write cat, duck, fish and butterfly.

An adult reads or explains as needed and uses intact, age-appropriate materials.

Try it together

1. Read: "I have wings. I also have feathers."
2. Find choices with wings, then check the second clue.
3. Explain why duck fits and butterfly does not; invent another clue pair.

Talk, draw or show your idea

Which clue distinguishes the two winged animals?

Space to explore

2 / 20 | Ages 5-6 | 15 minutes

Guide the robot on a grid

What we are practising

Plan and test a sequence of directional steps.

What you need

Paper / Pencil

An adult draws arrows while the child points.

Before you begin

Draw a 3-by-3 grid; start bottom-left and put a star top-right.

An adult reads or explains as needed and uses intact, age-appropriate materials.

Try it together

1. Use a finger as the robot; one arrow moves one square.
2. Plan four moves to the star without leaving the grid.
3. Follow the arrows exactly and change any mistaken instruction.

Talk, draw or show your idea

Can another four-move route reach the star?

Space to explore

3 / 20 | Ages 5-6 | 10 minutes

Find the broken pattern

What we are practising

Identify a repeating unit and check a sequence against it.

What you need

Paper / Pencil

Use claps and taps instead of marks.

Before you begin

Draw circle, square, triangle, circle, square, circle.

An adult reads or explains as needed and uses intact, age-appropriate materials.

Try it together

1. Explain that the first three shapes are the unit to repeat.
2. Read the sequence and stop where it breaks the rule.
3. Replace the sixth shape, then add the next three.

Talk, draw or show your idea

Which three shapes repeat as one group?

Space to explore

4 / 20 | Ages 5-6 | 15 minutes

Build from a spoken plan

What we are practising

Give precise instructions and compare models.

What you need

Six large blocks in two matching sets of three

Draw two matching sets of shapes instead.

Before you begin

Have the other builder face away while you arrange your set.

Use lightweight blocks on the floor and keep the model low.

Try it together

1. Place two blocks side by side with the third centred on top.
2. Describe the model without showing it and let the listener build.
3. Reveal both models, compare and improve one unclear instruction.

Talk, draw or show your idea

Which words explain where the third block belongs?

Space to explore

5 / 20 | Ages 5-6 | 10 minutes

Guess my sorting rule

What we are practising

Use examples and counterexamples to test a rule.

What you need

Paper / Pencil

Use large objects with wheels and without wheels.

Before you begin

Draw car and bicycle in a yes group; tree and cup in a no group.

An adult reads or explains as needed and uses intact, age-appropriate materials.

Try it together

1. Explain that one secret feature decides each group.
2. Invite “has wheels” as a possible rule and test a bus drawing.
3. Test a ball, which rolls but has no wheels, and discuss the rule.

Talk, draw or show your idea

Does everything that rolls have wheels?

Space to explore

6 / 20 | Ages 5-6 | 10 minutes

The toy train seats

What we are practising

Solve a position puzzle by elimination.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw three seats in a row labelled first, middle, last.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

- 1. Draw three seats in a row labelled first, middle, last.
- 2. Ask: Cat sits first. Bear sits last. Rabbit takes the remaining seat. Where is Rabbit?
- 3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw three seats in a row labelled first, middle, last. Cat sits first. Bear sits last. Rabbit takes the remaining seat. Where is Rabbit?

Space to explore

7 / 20 | Ages 5-6 | 10 minutes

Baskets with one label missing

What we are practising

Track used and unused options.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw three baskets labelled apples, bananas and blank; each holds a different one of apples, bananas, oranges.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw three baskets labelled apples, bananas and blank; each holds a different one of apples, bananas, oranges.
2. Ask: What label belongs on the remaining basket?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw three baskets labelled apples, bananas and blank; each holds a different one of apples, bananas, oranges. What label belongs on the remaining basket?

Space to explore

8 / 20 | Ages 5-6 | 10 minutes

The picnic before lunch

What we are practising

Order actions from explicit instructions.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw cards for eat, wash hands, put food on a plate.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw cards for eat, wash hands, put food on a plate.
2. Ask: Our rule is: wash before preparing the plate, then eat. What comes second?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw cards for eat, wash hands, put food on a plate. Our rule is: wash before preparing the plate, then eat. What comes second?

Space to explore

9 / 20 | Ages 5-6 | 10 minutes

Two doors, one key picture

What we are practising

Apply a symbol-matching rule.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a star on one door and a moon on another; the pretend key has a moon.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a star on one door and a moon on another; the pretend key has a moon.
2. Ask: This key opens only a door with the same symbol. Which door opens?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a star on one door and a moon on another; the pretend key has a moon. This key opens only a door with the same symbol. Which door opens?

Space to explore

10 / 20 | Ages 5-6 | 10 minutes

The no-circle collection

What we are practising

Understand a negative condition.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a circle, triangle, square and another circle.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a circle, triangle, square and another circle.
2. Ask: Collect every shape that is not a circle. Which two do you choose?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a circle, triangle, square and another circle. Collect every shape that is not a circle. Which two do you choose?

Space to explore

11 / 20 | Ages 5-6 | 10 minutes

A clue that helps

What we are practising

Compare how informative clues are.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a cat, dog and rabbit of the same size; give the rabbit long ears.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a cat, dog and rabbit of the same size; give the rabbit long ears.
2. Ask: Which clue singles out the rabbit here: “same size” or “long ears”?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a cat, dog and rabbit of the same size; give the rabbit long ears. Which clue singles out the rabbit here: “same size” or “long ears”?

Space to explore

12 / 20 | Ages 5-6 | 10 minutes

Turn-taking detective

What we are practising

Follow a repeating social rule.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Write or draw the fixed turn order: Lina, Omar, Lina, Omar.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Write or draw the fixed turn order: Lina, Omar, Lina, Omar.
2. Ask: Lina has just taken a turn. Under this alternating rule, whose turn is next?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Write or draw the fixed turn order: Lina, Omar, Lina, Omar. Lina has just taken a turn. Under this alternating rule, whose turn is next?

Space to explore

13 / 20 | Ages 5-6 | 10 minutes

The folded note

What we are practising

Distinguish a fold from a separation.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Fold a blank sheet once, open it and trace the crease.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Fold a blank sheet once, open it and trace the crease.
2. Ask: Did folding cut the paper into separate pieces?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Fold a blank sheet once, open it and trace the crease. Did folding cut the paper into separate pieces?

Space to explore

14 / 20 | Ages 5-6 | 10 minutes

The delivery that waits

What we are practising

Recognise a prerequisite.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a sealed pretend parcel and a box for its contents.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a sealed pretend parcel and a box for its contents.
2. Ask: Can we put the contents in the box before opening the parcel?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a sealed pretend parcel and a box for its contents. Can we put the contents in the box before opening the parcel?

Space to explore

15 / 20 | Ages 5-6 | 10 minutes

Can every animal win?

What we are practising

Use stated evidence instead of assumptions.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a race finish: Fox first, Hare second, Mouse third; define winner as first place only.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a race finish: Fox first, Hare second, Mouse third; define winner as first place only.
2. Ask: Under this rule, which animal wins?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a race finish: Fox first, Hare second, Mouse third; define winner as first place only. Under this rule, which animal wins?

Space to explore

16 / 20 | Ages 5-6 | 10 minutes

The empty corner

What we are practising

Complete a grid under a coverage rule.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a two-by-two board with a star in every corner except bottom-right.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a two-by-two board with a star in every corner except bottom-right.
2. Ask: Where must the fourth star go so every square has one?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a two-by-two board with a star in every corner except bottom-right. Where must the fourth star go so every square has one?

Space to explore

17 / 20 | Ages 5-6 | 10 minutes

Which question should we ask?

What we are practising

Design a discriminating question.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a large red ball and a small red ball.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a large red ball and a small red ball.
2. Ask: To find the large ball, is "Is it red?" enough? What should we ask instead?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a large red ball and a small red ball. To find the large ball, is "Is it red?" enough? What should we ask instead?

Space to explore

18 / 20 | Ages 5-6 | 10 minutes

The blue-or-striped rule

What we are practising

Apply an inclusive either-condition rule.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a blue plain hat, red striped hat and red plain hat.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a blue plain hat, red striped hat and red plain hat.
2. Ask: Hats may enter if blue OR striped. Which hats enter?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a blue plain hat, red striped hat and red plain hat. Hats may enter if blue OR striped. Which hats enter?

Space to explore

19 / 20 | Ages 5-6 | 10 minutes

Do the clues agree?

What we are practising

Notice contradictory conditions.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw one pretend toy described as “entirely red” and “entirely blue”, with red and blue distinct.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw one pretend toy described as “entirely red” and “entirely blue”, with red and blue distinct.
2. Ask: Can both descriptions fit the same toy at the same time?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw one pretend toy described as “entirely red” and “entirely blue”, with red and blue distinct. Can both descriptions fit the same toy at the same time?

Space to explore

20 / 20 | Ages 5-6 | 10 minutes

A place for the missing tile

What we are practising

Reconstruct by comparing corresponding positions.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a row: circle, triangle, square; copy it below with the middle tile blank.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

Try it together

1. Draw a row: circle, triangle, square; copy it below with the middle tile blank.
2. Ask: The lower row must be an exact copy. What belongs in the blank?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a row: circle, triangle, square; copy it below with the middle tile blank. The lower row must be an exact copy. What belongs in the blank?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Two clues, one animal

A little hint

Does a butterfly have feathers?

Answer / what to notice

Only the duck among these choices has both wings and feathers.

An answer must fit every clue, not just the first.

Make it easier

Offer only duck and fish.

Take it further

Dictate two clues with exactly one answer among the four.

Another day, away from the screen

Try two-clue guessing with household objects.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Guide the robot on a grid

A little hint

You need two moves up and two right.

Answer / what to notice

Up, up, right, right works; other orders of two ups and two rights also work.

Testing instructions one at a time reveals where a sequence needs repair.

Make it easier

Use one row of three squares.

Take it further

Block the middle square and find two routes avoiding it.

Another day, away from the screen

Draw new start and finish positions and exchange instructions.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Find the broken pattern

A little hint

Compare positions one and four, then two and five.

Answer / what to notice

The sixth should be a triangle; next come circle, square, triangle.

Once a repeating unit is agreed, its positions become predictable.

Make it easier

Use an alternating circle-square pattern.

Take it further

Make a four-part repeating unit with one error for an adult.

Another day, away from the screen

Design a border with the repaired pattern.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. Build from a spoken plan

A little hint

Use beside, on top and in the middle instead of “there”.

Answer / what to notice

Two blocks form the base and one sits centrally above them; discuss differences without blame.

Comparing models shows which instructions need more detail.

Make it easier

Keep the model visible during building.

Take it further

Use four blocks per set and invite a clarifying question.

Another day, away from the screen

Describe a drawing for someone else to reproduce.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. Guess my sorting rule

A little hint

Check the no examples as well as the yes examples.

Answer / what to notice

The intended rule is “has wheels”: bus is yes and ball is no.

New examples distinguish similar rules, such as rolling and having wheels.

Make it easier

State the rule and sort together.

Take it further

Invent a rule with two yes and two no examples.

Another day, away from the screen

Play another round with things worn on feet.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

6. The toy train seats

A little hint

Cross out the two occupied seats.

Answer / what to notice

Rabbit sits in the middle.

First and last are taken, leaving only middle.

Make it easier

Draw Cat and Bear in place.

Take it further

Swap Cat and Rabbit and describe all three positions.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

7. Baskets with one label missing

A little hint

Each fruit name is used once.

Answer / what to notice

Oranges.

Apples and bananas are already assigned; oranges is the unused name.

Make it easier

Draw one orange beside the possible labels.

Take it further

Add a fourth fruit and a fourth basket with a second missing label; explain what extra clue is needed.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

8. The picnic before lunch

A little hint

Follow the stated order, one action at a time.

Answer / what to notice

Put food on the plate.

Washing is first and eating last, so preparing the plate is between them.

Make it easier

Act out washing first.

Take it further

Add tidying after eating and retell all four actions.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

9. Two doors, one key picture

A little hint

Compare symbols, not door size.

Answer / what to notice

The moon door.

Moon matches moon under the game's rule.

Make it easier

Trace the moon on both pictures.

Take it further

Use keys with two symbols and require both to match.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

10. The no-circle collection

A little hint

“Not” excludes the named shape.

Answer / what to notice

The triangle and square.

Neither is a circle; both circles are excluded.

Make it easier

Cover the circles before choosing.

Take it further

Ask for shapes that are not squares and compare the collection.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

11. A clue that helps

A little hint

A useful clue leaves one choice in this drawing.

Answer / what to notice

Long ears.

All three share the size; only the drawn rabbit has long ears.

Make it easier

Point to the ears in each picture.

Take it further

Draw a long-eared dog too; invent a second clue.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

12. Turn-taking detective

A little hint

The two players alternate.

Answer / what to notice

Omar's turn.

Alternation gives each player a turn after the other.

Make it easier

Point to the two names while saying the order.

Take it further

Use a three-person order and predict the next two turns.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

13. The folded note

A little hint

Can you still lift the whole sheet together?

Answer / what to notice

No; it remains one connected sheet.

A crease changes the shape without making a cut.

Make it easier

Open and close the same fold slowly.

Take it further

Draw a cutting line instead; explain how cutting would change the result without using scissors.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

14. The delivery that waits

A little hint

The contents are still enclosed.

Answer / what to notice

No; first open the parcel, then move the contents.

Access to the contents is a necessary earlier step.

Make it easier

Mime opening the parcel.

Take it further

Add reading a note inside; identify two actions that both require opening first.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

15. Can every animal win?

A little hint

Use position, not the animal's usual speed.

Answer / what to notice

Fox.

Fox is first in the given result, regardless of expectations.

Make it easier

Circle the first position.

Take it further

Change the rule to “first two receive ribbons”; name both recipients.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

16. The empty corner

A little hint

Find the only square without a star.

Answer / what to notice

In the bottom-right square.

The other three squares already meet the one-star rule.

Make it easier

Shade the empty square lightly.

Take it further

Rotate the sheet and describe where the same empty square moves.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

17. Which question should we ask?

A little hint

Both choices are red.

Answer / what to notice

No. Ask "Is it the large ball?"

Colour does not separate these choices, but size does.

Make it easier

Say each ball's size aloud.

Take it further

Add a large blue ball and find two questions that identify each one.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

18. The blue-or-striped rule

A little hint

Meeting either condition is enough.

Answer / what to notice

The blue plain hat and red striped hat.

One meets the colour rule and the other meets the pattern rule.

Make it easier

Test each hat against the two words separately.

Take it further

Add a blue striped hat; explain why it also enters.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

19. Do the clues agree?

A little hint

Entirely means every part.

Answer / what to notice

No, under these descriptions.

Every part cannot be exclusively red and exclusively blue at once.

Make it easier

Compare two solid-colour drawings.

Take it further

Change “entirely” to “partly”; draw a toy that now fits both.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

20. A place for the missing tile

A little hint

Compare the same position in both rows.

Answer / what to notice

A triangle.

The middle position in the model contains a triangle.

Make it easier

Draw a line linking the two middle positions.

Take it further

Leave two blanks and restore both without changing their order.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.